

EasyShop – ShoppingTime

Shopping Time

Practical Activity Report

Submitted for
UI & UX SPECIALIST (UCS542)
End-Semester Lab Evaluation

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INDEX

1. INTRODUCTION
2. PROBLEM STATEMENT
3. SPECIFIC REQUIREMENTS
4. CONTEXT LEVEL DIAGRAM & DATA FLOW DIAGRAM
5. SYSTEM SPECIFICATION
6. TOOLS USED
7. SYSTEM DESIGN (UI/UX APPROACH)
8. FEATURES IMPLEMENTED
9. FIREBASE INTEGRATION DETAILS
10. SAMPLE SCREENSHOTS
11. OUTPUT REPORTS
12. TESTING & DEBUGGING
13. CONCLUSION

Introduction

EasyShop – ShoppingTime is a modern e-commerce web application designed to provide users with an intuitive and responsive shopping experience. The application leverages a component-based approach using React and serves as a frontend-focused project integrated with Firebase services for real-time data management. This report presents an in-depth analysis of the application's development, system design, architecture, and implementation details.

Problem Statement

Modern retail users expect seamless, fast, and mobile-friendly shopping experiences. However, many small-scale online shops still struggle with outdated interfaces, slow loading times, and lack of real-time inventory management. The problem identified was to design an aesthetically pleasing, fast, and highly interactive e-commerce platform that supports browsing, cart management, product filtering, and Firebase-backed data operations. The solution must focus on user experience, minimal load time, component reusability, and simplified data handling.

Objectives

- Provide a visually appealing and responsive UI for online shopping.
- Ensure smooth navigation between product categories.
- Implement a dynamic cart system with real-time updates.
- Use Firebase services to store products, cart data, and user selections.
- Enhance performance using React + Vite development environment.
- Ensure modular component structure for easy maintainability.

Specific Requirements

Functional Requirements:

1. Users should be able to browse all products across various categories.
2. Users must be able to add, remove, and update product quantities from the cart.
3. Firebase should store product data and images.
4. Application should load product galleries dynamically.
5. System should display individual product details, prices, and attributes.

Non-Functional Requirements:

1. High responsiveness across all devices.

2. Low latency achieved by Vite bundler.
3. Clean UI design aligned with modern e-commerce interfaces.
4. Component reusability and clean state management in React.

Tools and Technologies Used

1. React – For building a modular, component-based frontend.
2. Vite – Lightning-fast development server and build tool providing optimized bundling.
3. Firebase:
 - Firebase Firestore for scalable NoSQL product storage.
 - Firebase Hosting (optional) for deployment.
 - Firebase Storage for product images.
4. JavaScript – Core development language.
5. CSS3 – Styling with reusable classes and responsive layouts.
6. GitHub – Version control and collaborative repository management.
7. VS Code – Development IDE with integrated Git tools.

System Design

The EasyShop application follows a modular component-driven architecture. Core design elements:

1. Frontend Architecture:
 - Organized into pages (Landing, Product Listing, Product Detail, Cart, Checkout).
 - Shared components include navigation bars, card components, attribute selectors, and sliders.
2. Firebase Integration:
 - Firestore Database stores product catalogs with fields:
 - product_id

- name
- brand
- price
- gallery (array of image URLs)
- attributes (size, color options)
- Firebase Storage hosts images for optimized CDN delivery.

3. State Management:

- Local React state is used for cart and UI interactions.
- Cart updates propagate through the component tree.

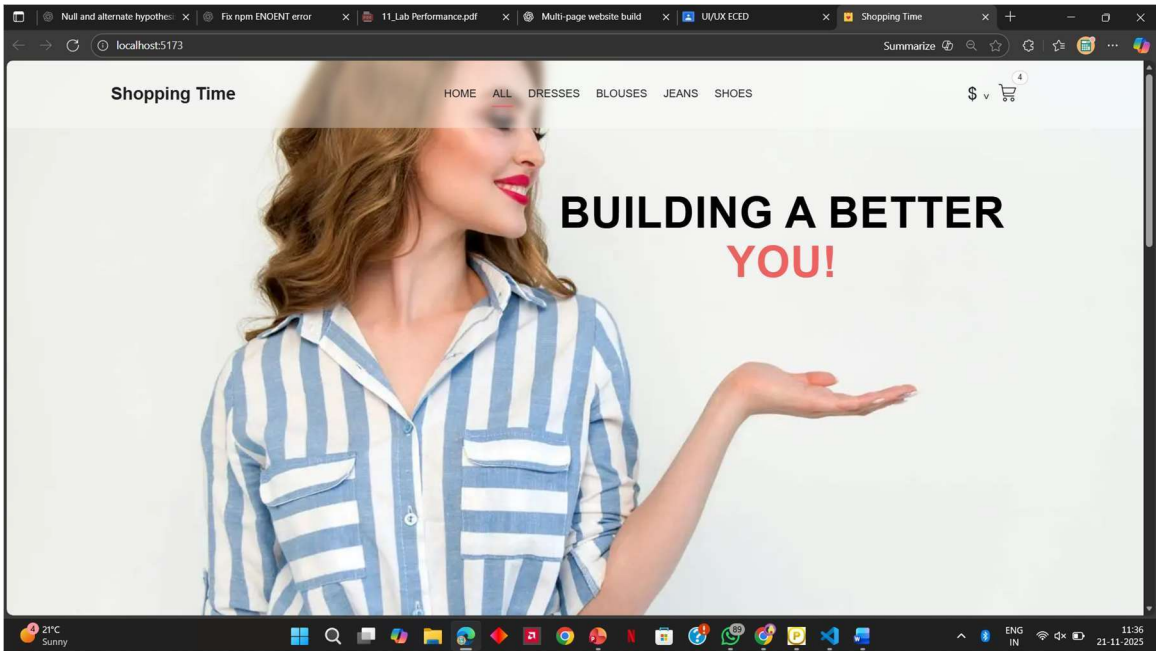
4. Routing:

- React Router DOM handles navigation from home to product pages.

5. Data Flow:

- Fetch products from Firebase → render categories → user selects product → cart updates dynamically.
- Checkout screen reads the final cart state to show total amount.

Output Screens (Placeholders)



Brand puff sleeve dress
\$79.00



H&M Summer Linen Dress
\$1999.00



ZARA
\$1199.00



My cart, 4 items

1

Abstract Print Blouse

\$0.00

2

puff sleeve dress

\$79.00

Total:

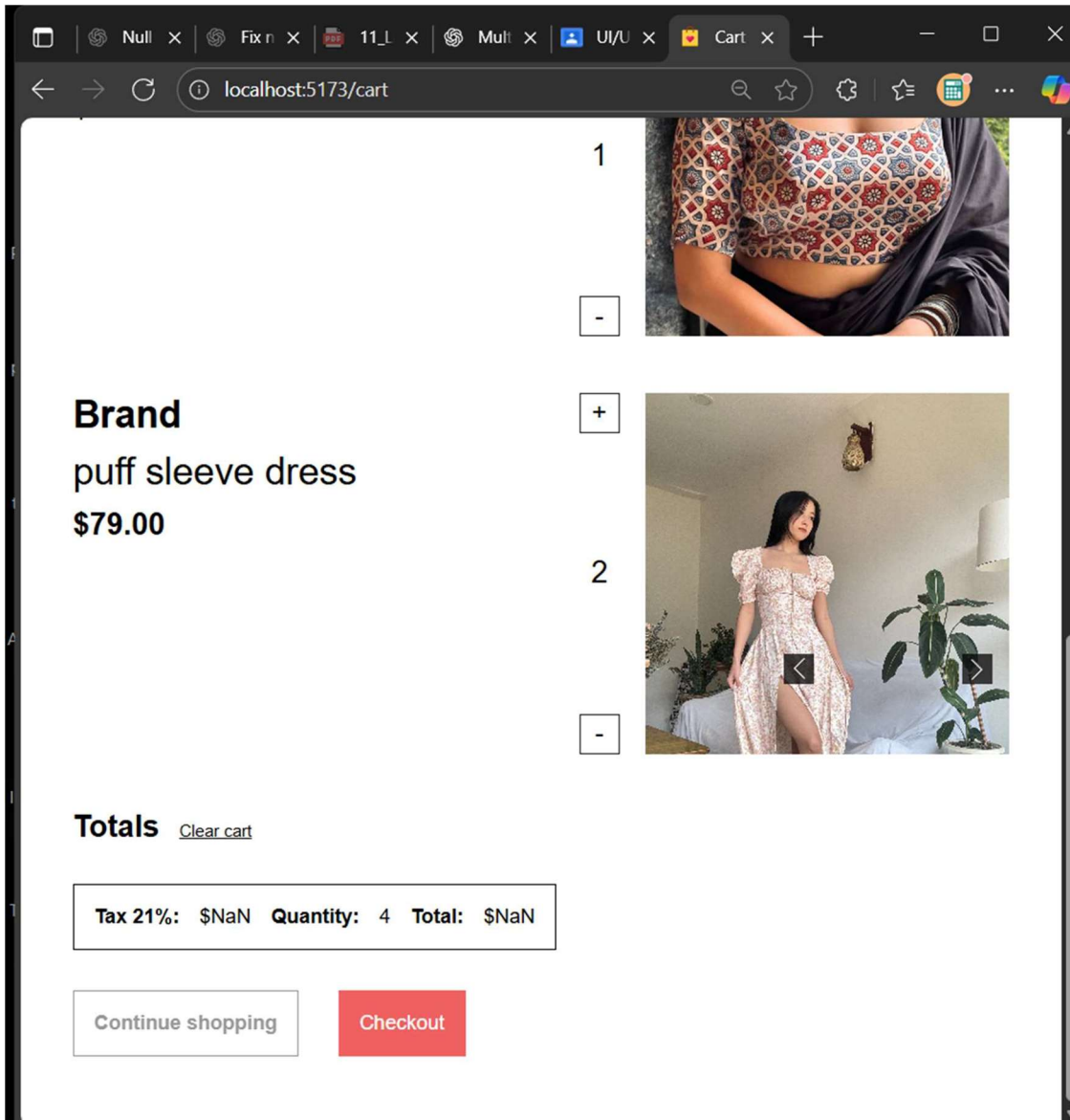
\$NaN

View cart

Clear cart

LET'S CREATE YOUR OWN STYLE

Dresses



Shopping Time

HOMEALLDRESSESBLOUSESJEANSSHOES

\$ v 4

Checkout

1

2

3

Personal detailsDelivery addressPayment method

Personal details

archi

aggarwal

architaggarwal661@gmail.com

06283257830

Continue to address

Shopping Time

HOMEALLDRESSESBLOUSESJEANSSHOES

\$ v 4

Checkout

1

2

3

Personal detailsDelivery addressPayment method

Delivery information

F-7/73 kashmir avenue

India

143001

Amritsar

Punjab

Continue to payment

Shopping Time

HOMEALLDRESSESBLOUSESJEANSSHOES

\$ v 4

Checkout

1

2

3

Personal detailsDelivery addressPayment method

Payment

Card number

MM / YY

CVV

Submit

Go back

Testing

Manual and functional testing were performed:

- Tested product loading from Firebase.
- Tested image slider functionality for multi-image products.
- Checked add/remove cart operations.
- Tested responsiveness on mobile, tablet, and desktop.
- Verified smooth navigation and component rendering.
- Checked handling of missing product fields to prevent crashes.

Conclusion

The EasyShop – ShoppingTime project demonstrates a successful implementation of a modern e-commerce web interface using React and Firebase integration. It effectively fulfills the objectives of creating a lightweight, fast, and user-friendly shopping system with modular design. Future improvements may include authentication, order history saving, payment gateway integration, and dashboard analytics.